The coefficient of static friction is equal to tangent of the angle of friction

GBV47

3.6.6 Angle of Repose

Consider an inclined plane on which an object is placed, as shown in Figure 3.29. Let the angle which this plane makes with the horizontal be θ. For small angles of θ, the object may not slide down. As θ is increased, for a particular value of θ, the object begins to slide down. This value is called angle of repose. Hence, the angle of repose is the angle of inclined plane with the horizontal such that an object placed on it begins to slide.

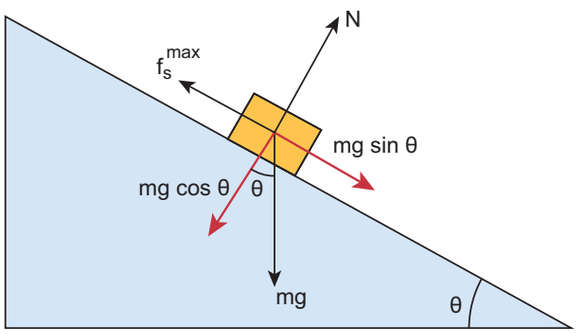


Figure 3.29 Angle of repose

Let us consider the various forces in action here. The gravitational force mg is resolved into components parallel ($mg\sin\theta$) and perpendicular ($mg\cos\theta$) to the inclined plane.

The component of force parallel to the inclined plane (mg sinθ) tries to move the object down.

The component of force perpendicular to the inclined plane (mg cosθ) is balanced by the Normal force (N).

$$N = mg\cos\theta$$

When the object just begins to move, the static friction attains its maximum value

$$f_s = f_s^{max} = \mu_s N = \mu_s mg\cos\theta \qquad (3.35)$$

This friction also satisfies the relation

$$f_s^{max} = mg\sin\theta \qquad (3.36)$$

Dividing equations (3.35) and (3.36), we get

$$\mu_s = \sin\theta / \cos\theta$$

From the definition of angle of friction, we also know that

$$\tan\theta = \mu_s, \qquad (3.37)$$

in which θ is the angle of friction.

Thus the angle of repose is the same as angle of friction. But the difference is that the angle of repose refers to inclined surfaces and the angle of friction is applicable to any type of surface.

UNIT-3(XI-Physics_Vol-1).indd 142 05-01-2022 22:18:21